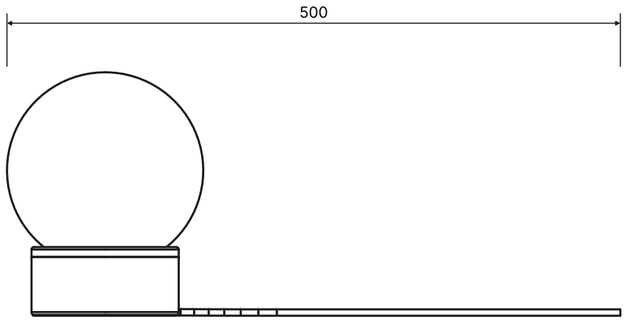
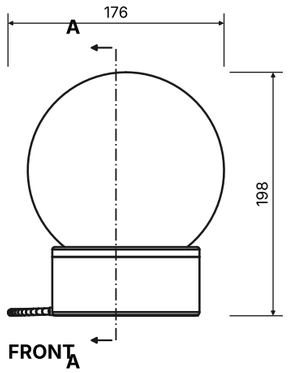
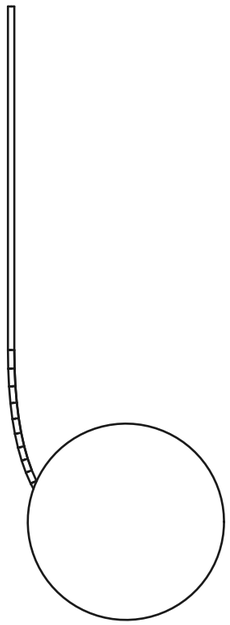
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.




FROM THE RIGHT




FRONT



FROM ABOVE

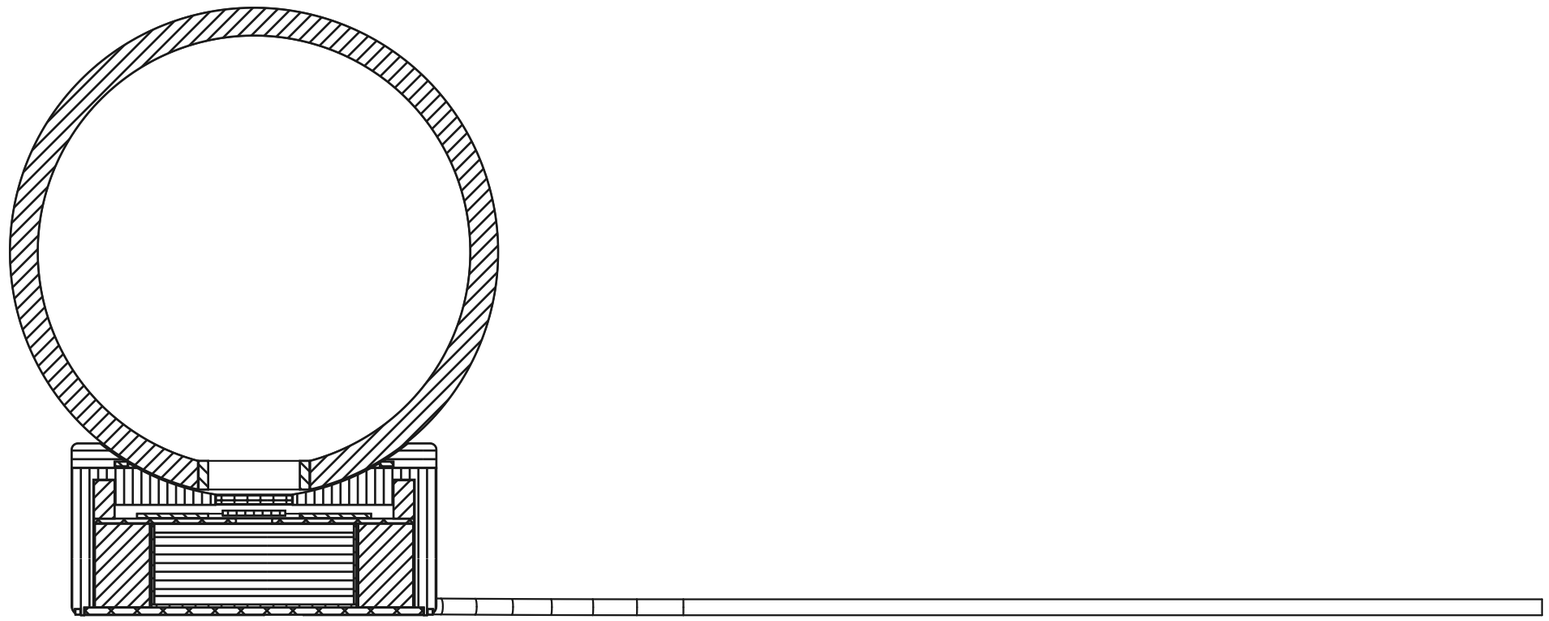
# General arrangement

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x -8.1 and seen from the right; cut faces hatched, a different angle for each part.



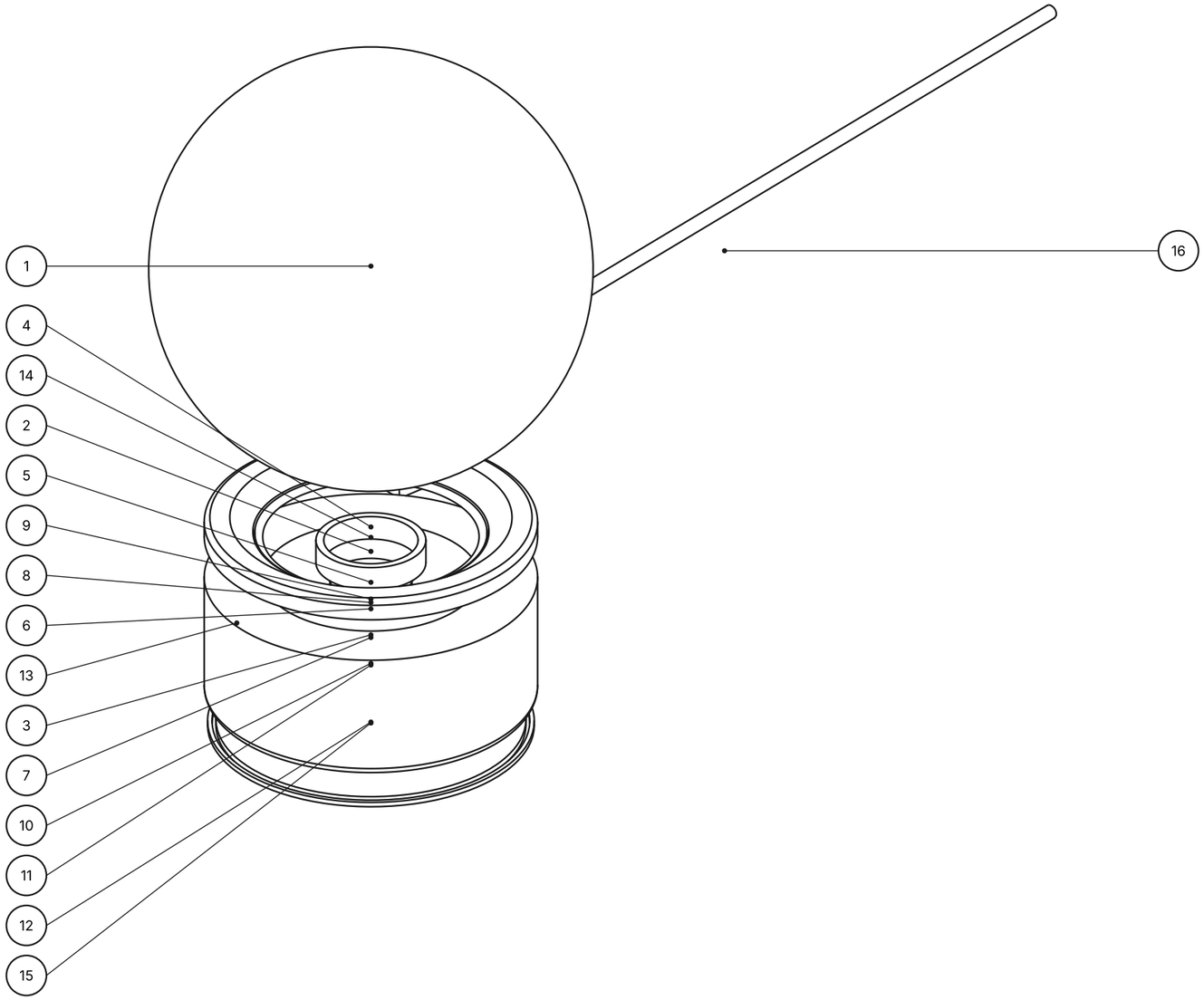
## Section A-A

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).




# Exploded view

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

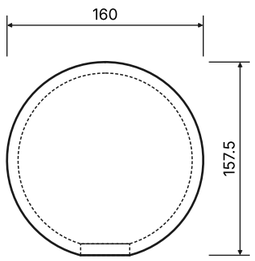
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

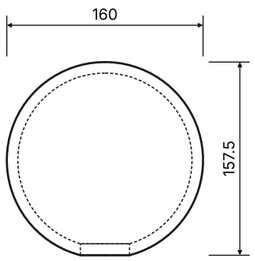
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**1. shell: formed, acrylic-formed; 634 cm3 each; 1 for the set; 1:5**

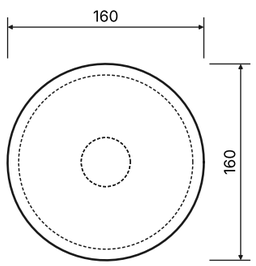
the ball: two hemispheres drape-formed from 10 mm tinted cast sheet on one aluminium mould (the production process; the prototypes are formed on it), their insides bead-blasted, joined at the equator with a polymerising acrylic cement (Acrifix 192 kind), the joint turned and polished: a faint line at the equator (to judge on the first sample); a 40 port cut at its underside. One part where there were three: the frost scatters because it faces air




**FRONT**



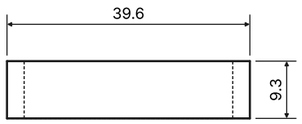

**FROM THE RIGHT**



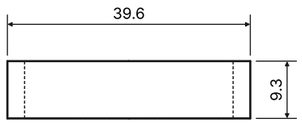

**FROM ABOVE**

**2. grommet: lathe, aluminium; 3 cm3 each; 1 for the set; 1:1**

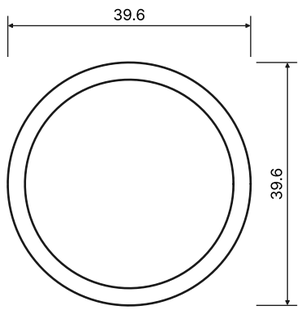
black anodised aluminium: lines the port, bonded




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

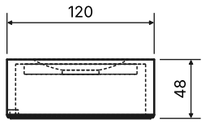
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

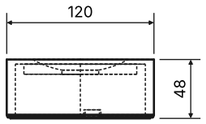
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**3. base: lathe, cast-acrylic; 176 cm3 each; 1 for the set; 1:5**

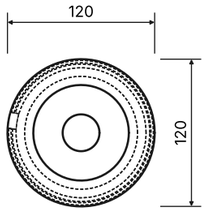
black cast acrylic, polished, turned from rod: 120 × 48 (was 30: the cells lie under the LED), the cup turned in it; from below the cells' pocket (106), the shoulder with threaded inserts for the chassis, the board's pocket (92), the ballast ring's groove, the hatch's counterbore (112) and the cable notch




FRONT



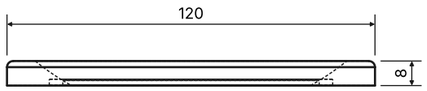

FROM THE RIGHT



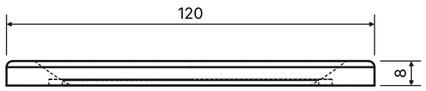

FROM ABOVE

**4. ring: lathe, aluminium; 36 cm3 each; 1 for the set; 1:2**

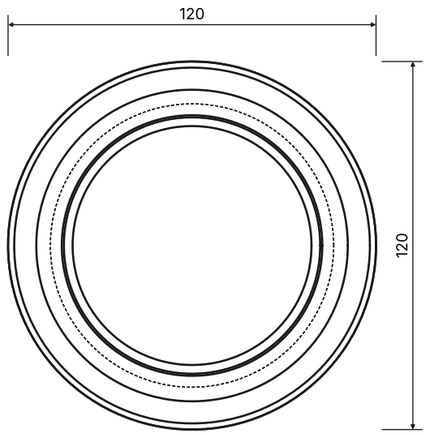
aluminium, black anodised, satin, insulated from the chassis: the touch (tap to switch, hold to dim, 2200 to 2700 K)




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

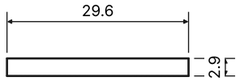
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

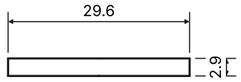
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**5. window: lathe, cast-acrylic; 2 cm3 each; 1 for the set; 1:1**

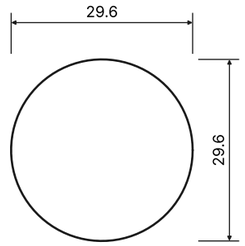
a clear acrylic plug, sealed with an O-ring: the LED under it, the base IP44 with the ball off




**FRONT**



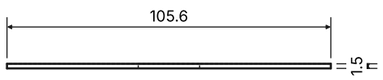

**FROM THE RIGHT**



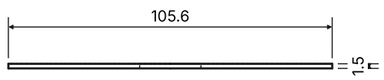

**FROM ABOVE**

**6. chassis: mill, aluminium; 13 cm3 each; 1 for the set; 1:2**

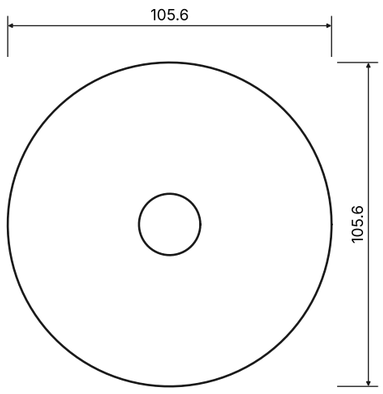
aluminium 6061 plate: the LED's heat path and the board's carrier, screwed up into the shoulder on threaded inserts, an O-ring between; it closes the cells' bay from above




**FRONT**




**FROM THE RIGHT**




**FROM ABOVE**

## Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

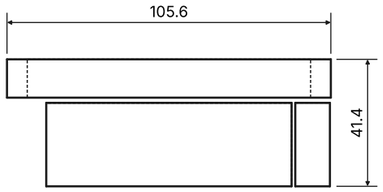
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

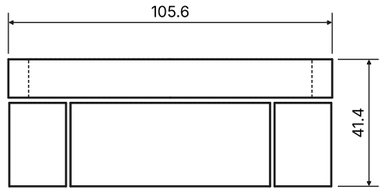
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

**7. ballast: mill, steel; 95 cm3 each; 1 for the set; 1:2**

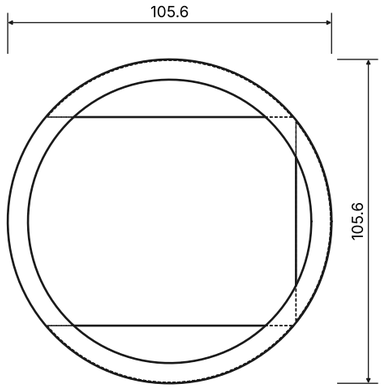
mild steel, zinc-plated: three segments round the cells' holder and a ring under the cup, screwed to the chassis (about 0.8 kg)




FRONT



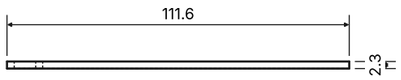

FROM THE RIGHT



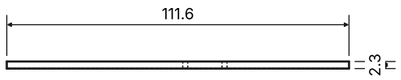

FROM ABOVE

**12. hatch: mill, aluminium; 22 cm3 each; 1 for the set; 1:2**

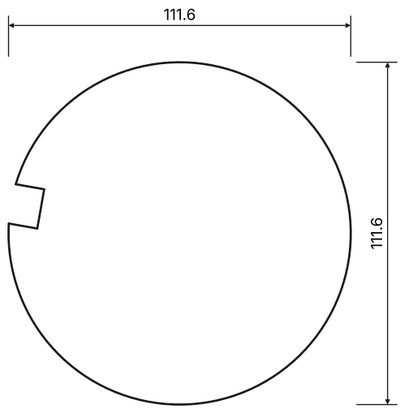
black anodised aluminium, 2.3 thick, in a counterbore under the base: four captive M3 screws (Phillips or Torx: commercially available tools), an O-ring (IP44); a label inside: the cells' type and how to change them




FRONT




FROM THE RIGHT




FROM ABOVE

## Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

| no. | item | make | qty | what | size | each USD |
|---|---|---|---|---|---|---|
| 1 | shell | formed | 1 | Cast acrylic sheet, tinted through, drape-formed over an aluminium mould | 634 cm3, 170 × 170 × 167 | 1,546.84 |
| 2 | grommet | lathe | 1 | Aluminium 6061 | 50 × 50 × 19 | 46.19 |
| 3 | base | lathe | 1 | Cast acrylic tube, tinted through | 130 × 130 × 58 | 124.21 |
| 4 | ring | lathe | 1 | Aluminium 6061 | 130 × 130 × 18 | 52.61 |
| 5 | window | lathe | 1 | Cast acrylic tube, tinted through | 40 × 40 × 13 | 45.81 |
| 6 | chassis | mill | 1 | Aluminium 6061 | 116 × 116 × 12 | 48.84 |
| 7 | ballast | mill | 1 | Mild steel | 116 × 116 × 51 | 51.87 |
| 8 | board | buy | 1 | to make (a contract PCB house) Core's board: a USB-C sink (5 V 3 A, no PD needed; a PD sink optional), a LiFePO4 charger with power path (TI BQ25171 kind, or Microchip MCP73123 kind: 3.6 V full, the lamp runs while it charges), a charge | | 28.00 |
| 9 | led | buy | 1 | Nichia or Seoul (to choose) A 30 mm round warm-dim LED board, two channels (2200 K and 2700 K, CRI 90+), about 2.5 W at full, 200 lm; a second 4-LED ring at the column's foot to light the column (0.3 W) | | 9.00 |
| 10 | cells | buy | 1 | any reputable LiFePO4 maker (to qualify: A123/Lithium Werks, EVE, Lishen) Three LiFePO4 26650 cells (IFR26650), about 3,000 mAh at 3.2 V each, in parallel (1S3P): about 29 Wh; about 2,000 cycles to 80 % (several times ordinary Li-ion's 500 or | | 12.00 |
| 11 | holder | buy | 1 | Keystone (or alike), to make A moulded holder for three 26650 cells with spring contacts, keyed for polarity, its NTC against the cells, wired to the board | | 5.00 |
| 12 | hatch | mill | 1 | Aluminium 6061 | 122 × 122 × 12 | 49.55 |
| 13 | usb | buy | 1 | Two EPDM O-rings (the window, the chassis to the drum), a silicone foot ring 110 x 2 (Shore 40, non-slip), a sealed USB-C receptacle (IP67 rated, e.g. GCT USB4135 kind): IP44 | | 5.00 |
| 14 | seat | buy | 1 | A silicone seat ring 92 × 84 × 1.5 (Shore 40, black) in the base's cup: the ball rests on it, and it stops a nudge rolling it | | 1.50 |
| 15 | foot-ring | buy | 1 | A silicone foot ring 117 × 114 × 2 (Shore 40, non-slip), and the hatch's O-ring | | 2.00 |
| 16 | cable | buy | 1 | any (made to order) A slim black braided-fabric USB-C to USB-C cable, 2 m, 3 A, a black right-angle overmoulded plug for the notch | | 6.00 |
| 17 | adapter | buy | 1 | any (UL/CE listed) A 20 W USB-C PD wall adapter, black, UL 62368-1 / CE listed (bought listed, so the plug-in Orb is a low-voltage lamp) | | 7.00 |
| 18 | screws | buy | 1 | Four captive M3 × 6 stainless pan-head screws (Phillips or Torx T10) with retaining washers, for the hatch | | 1.50 |

## Parts list

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

## Notes

1. Core (the Orb until round 35): a crystal ball, a 160 tinted ball low in the cup of a 120 × 48 base, 198 tall. The ball lifts off: the LED is under a sealed window in the base, so balls swap colour by hand.
2. One ball where there were three layers: tinted cast acrylic, two halves formed on one mould and joined at the equator, their inside bead-blasted. The frost faces air, so it scatters; seen from outside it is behind the tint, and the port that would show it faces the base.
3. Plugged in or not, one product: three LiFePO4 26650 cells (about 29 Wh) in a holder in the base, behind a hatch underneath on four captive screws, the user's to replace with standard cells. It runs from USB-C (5 V 3 A, the adapter and a 2 m fabric cable in the box) or from its cells, and charges while it runs, in about 3.5 hours. About 9 hours at full brightness (2.5 W) at the default charge limit (90 %), about 10.5 at a full charge.
4. Long life: LiFePO4 keeps about 80 % of its capacity after about 2,000 cycles (ordinary Li-ion about 500); the charge limit, a battery-management circuit (protection, a fuel gauge, charging held off below 0 C and above 45 C by the cells' NTC) and no exposed contacts. Certified as one product: IEC/UL 62368-1 with its battery annex; the cells to IEC 62133-2 and UN 38.3.
5. Touch the black ring: tap to switch, hold to dim, 2200 to 2700 K as it dims. IP44: the base sealed by O-rings at the window, the chassis and the hatch, the USB-C receptacle sealed; with the ball on or off.
6. Black, so the tint is the only colour: the base black cast acrylic, polished, over an aluminium chassis and a steel ballast; the ring, the port's lining and the hatch black anodised aluminium.

## Notes

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15